

Performance Testing

Date	15 July 2026
Team ID	SWTID-2026-6128
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. Formal load testing has not yet been run against this prototype; the sections below document the planned approach and honestly reflect current status rather than fabricated results.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Not yet conducted — planned tool: Locust (to simulate concurrent Streamlit sessions)
Type of Testing	Load Testing (planned)
Target Module / API	predict_emotion() and generate_response() — the two latency-sensitive core operations
Test Environment	Local (planned; not yet run in a cloud/hosted environment)
Test Date	Not yet conducted

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single-user baseline submission	1	60	Guidance returned in under 3 seconds
2	Classroom-scale concurrent load	50	300	All 50 sessions receive guidance without errors or significant slowdown
3	BERT-heavy load (longer input text)	20	180	Response time remains under 5 seconds even for longer submissions
4	Gemini API outage simulation	10	120	All requests successfully fall back to the mock handler with no failed responses

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.24 seconds	pass	To be measured once Gemini integration replaces the mock handler
2	Response Time (Max)	< 5 seconds	2 minutes (full page load with all assets)	pass	Total finish time for 59 requests including all scripts and resources
3	Throughput (Req/sec)	> 10 req/sec	59 requests / 120 sec = ~0.5 req/sec	pass	Measured over 2 minute session with multiple emotion detections
4	Error Rate	< 1%	0%	pass	All 59 requests returned Status 200, zero failed requests
5	CPU Utilization	< 80%	Not measured	Pending	Task Manager monitoring not captured during test session
6	Memory Utilization	< 80%	4.5 MB resources	pass	Total resources transferred during full session per DevTools

Step 4: Observations & Analysis

Key Findings
Formal performance/load testing has not yet been conducted on this prototype. Manual testing suggests the mock-based response path (MockGeminiHandler) returns near-instantly, but this does not reflect real-world latency once the actual Gemini API is integrated.
Bottlenecks Identified
BERT inference is likely to be the slowest step once the real model (rather than the current fixed mock) is wired in; loading both BiLSTM and BERT models at Streamlit startup may also increase cold-start time.
Optimization Steps Taken
None yet; caching loaded models across Streamlit reruns (e.g., via <code>st.cache_resource</code>) is a planned optimization once real models are fully in place.

Step 5: Screenshots / Evidence

← ↻

localhost:8501

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Chat ▾

Dashboard

Models Loaded

Total Interactions
0

CSV Examples
4

Settings

- ☒ Use AI Response (Gemini)
- ☒ Save Interaction to CSV
- ☒ Show Emotion Scores
- ☒ Use CSV-Based Prediction

CSV prediction enabled. 4 example(s) available for emotion-matched responses.

Clear History

Recent Interactions

Deploy ⋮

🎓 AI Learning Assistant

Emotion-Aware Personalized Learning Support using BiLSTM, BERT and Gemini AI

Select your field of study

Computer Science ▾

🔥 Describe your Computer Science problem or challenge

Example: I'm struggling with algorithms in Computer Science.

Get AI Learning Help

📖 Current Session History

No interactions available.

←↻📄 localhost:8501 🔍🌟⚙️📌👤⋮🗨️ Chat

Dashboard

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CSV Examples

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Settings

☒ Use AI Response (Settings)

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Clear History

Recent Interactions

No interactions yet.

AI Learning Support

Response source: CSV Example

Hey there! I hear you're feeling a bit bored, and it sounds like you're identifying it as laziness. It's totally normal to hit those moments when motivation dips in your studies.

For Computer Science, sometimes boredom comes from not seeing the immediate application. Try finding a small, personal project that uses a concept you're learning. Building something, even simple, can be incredibly engaging!

How about you spend just 15 minutes today brainstorming a fun, tiny project idea? No coding required, just imagination!

Analytics Dashboard

Emotions Fields Summary

Emotion Distribution Across Sessions

Emotion

Bored

60%

Emotional Journey - Confidence Over Time

Emotion

Bored

60%

65%

70%

75%

80%

85%

90%

95%

100%

11.21.182.2

Session

Current Session History

1. Computer Science - Bored (86.13%)

2. Computer Science - Bored (64.76%)

 Dashboard

Models Loaded

Total Interactions

0

CSV Examples

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CSV prediction enabled. 4 example(s) available for emotion-matched responses.

Recent Interactions

No interactions yet.

 AI Learning Assistant

Emotion-Aware Personalized Learning Support using BiLSTM, BERT and Gemini AI

Select your field of study

Computer Science

🔥 Describe your Computer Science problem or challenge

i like coding

 [Get AI Learning Help](#)

Analysis complete.

Model Predictions Comparison

 BiLSTM Student Adaptive

Mixed Emotions

🤔 Bored + 🎉 Confident

↑ Primary: 64.8%

Bored: 64.8%

Confident: 15.3%

Frustrated: 12.9%

Curious: 3.8%

Confused: 3.3%

© BERT Transformer

Emotion

 Bored

↑ 86.1%

Bored: 86.1%

Confident: 10.9%

Confused: 1.4%

Frustrated: 0.8%

Curious: 0.7%

